

LESSON 4.8

EXPLORING THE AREA OF A CIRCLE



LESSON INTRODUCTION

MA.7.GR.1.4 Explore and apply a formula to find the area of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can use the area of a rectangle to develop the formula for the area of a circle.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How is the formula for the area of a parallelogram related to the formula for the area of a circle?
- What is the formula for the area of a circle?
- How is the circumference of a circle related to its area?

LESSON NARRATIVE

In this lesson, students decompose circles into sectors and rearrange the sectors so that the circle resembles a parallelogram. Students then use the formula for the area of a parallelogram to determine that the area is equal to half of the circle's circumference multiplied by its radius. After substituting the formula for the circumference, students derive a formula for the area of a circle.

COMPONENT SUMMARY

4.8.1 WARM-UP (~5 MIN.)

Decide whether the circular or rectangular table has a larger area

4.8.3 ACTIVITY (25 MIN.)

Discover the formula for the area of a circle

4.8.5 WRAP-UP (~5 MIN.)

Reflect on interest in and understanding of the lesson's learning target

4.8.2 GUIDED PRACTICE (5 MIN.)

Explore the definition of area

4.8.4 BRING IT TOGETHER (5 MIN.)

Engage in guided reflection and summary

RESOURCES

GLOSSARY

Key Terms:

- Area

Additional Terms:

- N/A

MATERIALS

- Circles separated into 4, 8, and 16 sections
- Different colored writing utensils

ADDITIONAL PREPARATION

- For Activity 4.8.3, make enough copies of the 3 circles for each set of partners.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have difficulty correctly making the series of substitutions needed to use the formula for the area of a rectangle to derive the area of a circle

THINGS TO CONSIDER

- Consider providing students with colored writing utensils. Prompt students to use a different color for each dimension and continue using that color throughout the series of substitutions.
- Consider creating an anchor chart or adding the information to a class anchor chart.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 4.8.4 Bring It Together, prompt students to compare and connect the formulas for the areas of parallelograms and areas of circles. Ask students to discuss the similarities and differences between the shapes and formulas.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Students use structure to make sense of the relationship between the area of a circle and the area of a parallelogram by decomposing a circle to resemble the shape of a parallelogram or rectangle. In doing this, students will begin making use of the structure by connecting the relationship between the radius in a circle and the height of a parallelogram as well as the relationship between the circumference and the base of the parallelogram.

DIFFERENTIATE INSTRUCTION

GeoGebra offers interactive, digital applets that students can explore while making connections between the area of a circle and triangles.

For example, teachers could project a GeoGebra applet of their choosing for the whole class to launch the Activity, having students make conjectures about the formula for area.

4.8.1 WARM-UP: WOULD YOU RATHER?

TEACHER MOVES

Notice that the Warm-Up is intentionally laying the foundation for making connections between the two shapes as students will be relating circles and parallelograms in the next activity.

Monitor for student responses that include decomposing the circle and comparing those two shapes. Take note of those responses from the whole group to use as connections for the next activity.

DIFFERENTIATE INSTRUCTION

Prompt students to sketch each shape in order to visualize each table. In addition, have the students compare and contrast each table.



4.8.1 Warm-Up: Would You Rather?

Table A has a length of 96 inches and a width of 42 inches.
Table B has a radius of 45 inches.

- Which table would you rather use for a large gathering?
Explain your choice using mathematics or a picture.

Answers may vary.

Sample answer:

Table A:

$$P = 2(l + w)$$

$$P = 2(96 + 42) = 276 \text{ in}$$

Table B:

$$C = 2\pi r$$

$$C = 2(3.14)(45) = 282.6 \text{ in}$$

Since Table B has a larger circumference, I think I could fit 1-2 more people at the gathering.



4.8.2 Guided Practice: Area of a Circle

A closed figure is an enclosed shape that occupies space. A circle is an example of a closed figure.

1. Draw a picture of a figure that is not closed.

Answers will vary.

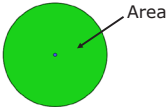


Area is the measure, in square units, of the inside region of a closed two-dimensional figure.

2. Discuss with your partner why a closed figure can have an area, but a figure that is open cannot have an area. Summarize your discussion.

A closed figure can have an area because all of its space is enclosed in a figure. An open figure cannot have an area because the space is not bound within the figure.

The measure of the space inside the circle is the circle's area.



4.8.2 GUIDED PRACTICE: AREA OF A CIRCLE

QUESTIONING

Consider asking students to think of a figure in the classroom for which they can find the area, and a figure in the classroom for which they cannot find the area.



4.8.3 Activity: The Formula for the Area of a Circle

1. Your teacher will give you and your partners each a circle that is separated into four, eight, or sixteen sections.
- Draw an image of your circle and your partners' circles in the first column of the table.
 - Cut out each of the sections of your circle.
 - Rearrange your sections, side by side, to compose a figure that resembles a parallelogram. Paste this in the last column.
 - Draw an image of both of your partners' composed shapes in the last column.

	Circle	Parallelogram
Circle with 4 Sections		
Circle with 8 Sections		
Circle with 16 Sections		

4.8.3 ACTIVITY: THE FORMULA FOR AREA OF A CIRCLE

TEACHER MOVES

Compare and Connect

Ask students to compare the parallelograms. Is there a relationship between the number of sectors and how much the rearranged sectors resemble a parallelogram?

DIFFERENTIATE INSTRUCTION

Students begin with a blank slate. If students are unsure how to start, consider modeling one example.

4.8.3 ACTIVITY: THE FORMULA FOR AREA OF A CIRCLE (CONTINUED)

ENRICHMENT

Why do you think the circle with the most sections more closely resembles a rectangle than a parallelogram?
Use this question, if needed, to set up the next part of the activity.

TEACHER MOVES

Think-Pair-Share

Students can either discuss then summarize their discussion (as seen in question 3), or students can write their responses independently first, then share their responses with a partner before editing their responses based on their discussion.

TEACHER MOVES

Clarify that in question 4 students should determine the number of sectors only along the base, not the total number of sectors.

2. Complete the statements.
The circle with 4 sections resembles a

☐ square.
☒ parallelogram.

The circle with 8 sections resembles a(n)

☒ parallelogram.
☐ octagon.

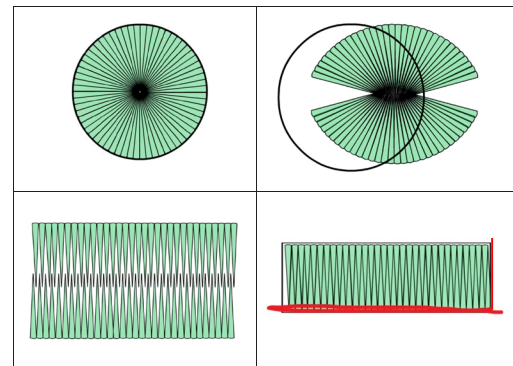
The circle with 16 sections most closely resembles a

☐ kite.
☐ square.
☒ rectangle.

3. Discuss with your partners what shape you will see if the number of sectors continues to increase. Summarize your discussion.

As the number of sectors increases, the shape will become more like a rectangle.

This circle has been divided into 64 equal sectors. The sectors were then separated in half and fit together to compose a figure as shown.

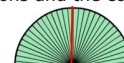


4. How many sectors make up the base of the composed shape in the last figure?
32

5. Trace the base of the composed shape and its corresponding part of the original circle.



6. Trace the height of the shape composed with the 64 sections and the corresponding part of the circle.



7. Complete the following statements.

The formula for the area of a rectangle is

$A = b \times h.$

$A = 2\pi r.$

The dimension of the circle that corresponds to one height of the rectangle is

the radius.

half the circumference.

The dimension of the circle that corresponds to one base of the rectangle is

the radius.

half the circumference.

8. Given the formula $C = 2\pi r$, work with your partner to find the measure of half the distance around a circle.



$C = 2\pi r$	Formula for the circumference of a circle.
$C = 2\pi r$ 2 2	Divide both sides of the equation by 2.
$\frac{1}{2}C = \pi r$	Equation written in simplified form for half of the circumference.

9. Given the formula for the area of a rectangle, substitute the corresponding parts of the circle to discover the area of a circle.

$A = b \times h$	Formula for the area of a rectangle.
$A = \pi r \times r$	Substitute the parts of the circle that correspond with the base and height of the rectangle.
$A = \pi r^2$	Multiply like bases to write the simplified formula of a circle.

4.8.3 ACTIVITY: THE FORMULA FOR AREA OF A CIRCLE (CONTINUED)

TEACHER MOVES

Sentence frames are included in this activity as scaffolding for students to engage independently or with a partner.

TEACHER MOVES

Students can refer to their responses from question 7 if they are unsure what to substitute.

4.8.4 BRING IT TOGETHER: AREA OF A CIRCLE

TEACHER MOVES

Stronger and Clearer

Provide students with an opportunity for synthesis to formalize their findings from the lesson. Prompt students to make sense of their fill-in-the-blank responses using their own words. Student should share with a partner, receive feedback, then implement that feedback for a more precise, clearer response.

TEACHER MOVES

The formula, $A = \frac{1}{2}Cr$, is not one that is often used in textbooks. It was derived for in this lesson to extend a student's thinking and understanding of the connectiveness found in mathematics. In this lesson, students were exposed to mathematical thinking that included pondering an infinite number of sectors and justifying formulas using other formulas. Adding the formula and a visual to an anchor chart can help students remember their thinking processes.



4.8.4 Bring it Together: Area of a Circle

1. Use the words or phrases from the box to complete the sentences.

height	rectangle	half of the circumference
circle	radius	sector
		circumference

The formula for the area of a rectangle was used to discover the area of a circle.

When a circle is divided into sectors, the sectors can be arranged into a rectangle. The base of the rectangle is half of the circumference of the circle. The height of the rectangle is the radius of the

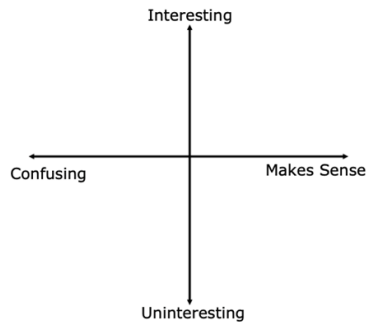
circle. Using this method results in the formula, $A = \frac{1}{2}Cr$.



4.8.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest in today's learning target and how much it made sense to you.

I can use the area of the rectangle to develop the formula for the area of a circle.



Answers will vary.

4.8.5 WRAP-UP: REFLECTION

TEACHER MOVES

Prompt students to include coordinates in their responses for more details. Have students explain why they chose each coordinate based on the learning target for this lesson.

For example, students may say, "I chose (5,1) because I thought it was cool how you could break a shape down (decompose) to find another shape, but I still have some questions on finding area in word problems."